

CampusConnect

A Smart Campus Exchange Platform

Overview

This document presents the project execution plan for CampusConnect, covering the team-level timeline and milestones, module ownership and task assignments, individual execution plans and checkpoints, cross-module dependencies, and the tools used to plan and track work. It builds on the scope, technology stack, and roles defined in the Project Proposal, and translates them into a week-by-week execution plan across the 14-week development cycle.

Team Task Timeline & Milestones

The team-level timeline is organized into eight phases, each with a defined duration and set of deliverables. Phases overlap where backend, frontend, and AI integration work can proceed in parallel, allowing module leads to begin dependent tasks as soon as the required APIs or designs are available.

Phase	Milestone	Duration	Key Deliverables
Phase 1	Planning & Requirement Analysis	Week 1 - 2	Finalized scope, requirements gathering, wireframing, and technology setup.
Phase 2	UI/UX Design	Week 3 - 4	Design system, wireframes to high-fidelity mockups, and design review.
Phase 3	Core Backend Development	Week 5 - 7	Database schema, authentication, listing CRUD APIs, and admin APIs.
Phase 4	Core Frontend Development	Week 6 - 9	React components for browsing, listing creation, chat UI, and dashboards.
Phase 5	AI Feature Integration	Week 8 - 10	AI-assisted listing creation, intelligent matching engine, and OCR duplicate detection.
Phase 6	Real-Time Features & Notifications	Week 10 - 11	In-app chat, real-time notifications, and rental reminders via Socket.io.
Phase 7	Testing & QA	Week 12 - 13	Unit testing, integration testing, bug fixing, and security checks.
Phase 8	Deployment & Final Presentation	Week 14	Deployment to cloud, documentation, and final project demo.

The following milestones mark the completion of each major capability and are used to track whether the team remains on schedule:

- Requirements, architecture, and technology stack finalized — end of Week 2.
- High-fidelity UI/UX design approved — end of Week 4.
- Core backend APIs (authentication, listings, admin) live — end of Week 7.
- Core frontend integrated with backend services — end of Week 9.
- AI-assisted listing creation and matching engine integrated — end of Week 10.
- Real-time chat and notification system live — end of Week 11.
- Testing and QA sign-off across all modules — end of Week 13.
- Final deployment and project demonstration — Week 14.

Team Task Assignments

Each module has one accountable owner. In addition, every member is a named deputy for one other module that depends closely on their own work, so no module relies on a single point of contact if its owner is temporarily blocked or unavailable.

Team Member	Module Owned	Deputy / Support For
Dev Barot	Project Lead & System Architecture	Overall integration checkpoints across all modules
Dhrumil Doshi	Authentication & User Management	Kajal Varlani — admin access control & role checks
Nishka Shah	Marketplace & Listing Management	Charmi Bhayani — listing-to-rental workflow continuity
Heer Mehta	AI & Intelligent Services	Nishka Shah — AI-assisted listing features
Heer Shah	Search, Chat & Notification	Heer Mehta — match-event notification triggers
Charmi Bhayani	Rental & Transaction Management	Heer Shah — rental reminder notifications
Kajal Varlani	Admin Panel & Analytics	Deep Boghara — analytics data pipeline
Deep Boghara	Database, DevOps & Quality Assurance	Dev Barot — architecture & deployment decisions

Individual Team Member Task Assignments

Dev Barot (202512069) — Project Manager & System Architect

Phase	Weeks	Task
Planning	W1-2	Finalize requirements, define system architecture, and set up Git repository and project task board.
Design	W3-4	Review UI/UX design deliverables and validate architecture against approved wireframes.
Backend	W5-9	Coordinate backend and frontend integration; run sprint reviews with module leads.
Backend	W7	Run a mid-project architecture review across all module leads and update the project risk log.
AI	W8-10	Oversee integration of AI services into the core system architecture.
Real-Time	W10-11	Coordinate rollout of real-time features across all modules.
Testing	W11-13	Lead testing coordination and track bug resolution across teams.
Deployment	W14	Oversee deployment and prepare final documentation and presentation.

Dhrumil Doshi (202512088) — Authentication & User Management Lead

Phase	Weeks	Task
Planning	W1-2	Gather requirements for authentication, user roles, and access levels.
Design	W2-3	Design login, signup, and profile UI wireframes.
Backend	W3-5	Develop JWT-based authentication, password handling, and role-based access control APIs.
Frontend	W5-6	Build user profile and account settings frontend components.
Backend	W7-9	Support marketplace and rental modules with auth checks and access-control integration (deputy for Kajal Varlani).
Real-Time	W9-10	Integrate the ratings and reputation system with notification triggers.
Testing	W11-12	Test authentication flows and resolve security issues.
Deployment	W14	Support deployment of authentication services.

Nishka Shah (202512035) — Marketplace & Listing Management Lead

Phase	Weeks	Task
Planning	W1-2	Define item categories and listing workflow requirements.
Design	W3-4	Design listing creation and browsing wireframes.
Backend	W4-7	Develop listing CRUD APIs, category management, and image upload backend.
Frontend	W6-9	Build listing creation, browsing, and category UI components.
Backend	W8-9	Coordinate with Charmi Bhayani so listings transition cleanly into rental workflows (deputy pairing).
AI	W9-11	Integrate AI-assisted listing suggestions into the listing creation flow.
Testing	W11-13	Test buy, sell, rent, exchange, and giveaway workflows end-to-end.
Deployment	W14	Finalize the listing module for deployment.

Heer Mehta (202512095) — AI & Intelligent Services Lead

Phase	Weeks	Task
Planning	W1-3	Research AI and OCR requirements and evaluate Groq API feasibility.
Backend	W4-6	Set up the FastAPI AI service scaffolding, once early backend models are available.
AI	W6-10	Develop AI-assisted listing creation, price estimation, semantic matching engine, and OCR-based duplicate detection.
AI	W9-10	Support Nishka Shah in refining AI-assisted listing suggestions (deputy pairing).
Real-Time	W10-11	Integrate match-event triggers with the Redis pub/sub notification pipeline.
Testing	W10-11	Prepare an evaluation dataset and benchmark report for AI matching precision.
Testing	W11-13	Test AI recommendation accuracy and matching precision.
Deployment	W14	Finalize AI service deployment.

Heer Shah (202512024) — Search, Chat & Notification Lead

Phase	Weeks	Task
Planning	W1-2	Define search, filtering, and notification requirements.
Design	W3-4	Design chat and notification UI wireframes.
Backend	W4-7	Support backend API design for search and filtering endpoints.
Frontend	W7-9	Build advanced search, filters, saved searches, and wishlist UI.
Real-Time	W10-11	Integrate match-event triggers from Heer Mehta and rental due-date triggers from Charmi Bhayani into the notification pipeline.
Real-Time	W10-12	Implement Socket.io-based in-app chat and real-time notifications.
Testing	W12-13	Test chat reliability and notification delivery.
Deployment	W14	Finalize search, chat, and notification services for deployment.

Charmi Bhayani (202512028) — Rental & Transaction Management Lead

Phase	Weeks	Task
Planning	W1-2	Define rental and transaction management requirements.
Design	W3-4	Design rental request and QR handover wireframes.
Backend	W5-8	Develop rental request APIs and transaction state management.
Backend	W7-8	Integrate QR code generation and verification module.
Frontend	W7-9	Build the rental dashboard and due-date tracking UI.
Real-Time	W10-11	Implement rental reminder scheduling using node-cron/BullMQ.
Testing	W12-13	Test rental and transaction workflows end-to-end.
Deployment	W14	Finalize the rental module for deployment.

Kajal Varlani (202512017) — Admin Panel & Analytics Lead

Phase	Weeks	Task
Planning	W1-2	Define admin requirements and moderation workflows.
Design	W3-4	Design admin dashboard wireframes.
Backend	W7-9	Develop admin APIs for listing approval, user management, and dispute handling, once other modules generate real data to moderate.
Frontend	W9-11	Build the admin dashboard UI and analytics views.
Backend	W10-11	Coordinate with Deep Boghara on the analytics data pipeline feeding the admin dashboard (deputy pairing).
Real-Time	W11-12	Integrate reporting tools and moderation notifications.
Testing	W12-13	Test admin workflows and dispute resolution processes.
Deployment	W14	Finalize the admin panel for deployment.

Deep Boghara (202512076) — Database, DevOps & Quality Assurance Lead

Phase	Weeks	Task
Planning	W1-2	Design the database schema and set up MongoDB Atlas and the Git workflow.
Design	W3-4	Review schema design against approved UI/UX flows.
Backend	W3-4	Set up the CI pipeline and automated build checks ahead of active development.
Backend	W4-11	Support backend and frontend integration and manage version control across the team on an ongoing basis.
Deployment	W9-12	Monitor performance of AI and real-time services and prepare the deployment pipeline.
Testing	W12-13	Lead unit testing, integration testing, and security checks.
Deployment	W13	Support Dev Barot with a final architecture and release-readiness review before deployment (deputy pairing).
Deployment	W14	Manage deployment across Vercel/Netlify, Render/AWS, and MongoDB Atlas.

Individual Milestones

Team Member	Key Milestone	Target Week
Dev Barot	Architecture finalized and project board set up	Week 2
Dev Barot	Backend and frontend integration coordinated across modules	Week 9
Dev Barot	Final deployment and project demonstration delivered	Week 14
Dhrumil Doshi	Authentication and RBAC APIs live	Week 5
Dhrumil Doshi	Ratings and reputation system integrated	Week 10
Dhrumil Doshi	Authentication module fully tested	Week 12
Nishka Shah	Listing CRUD APIs live	Week 7
Nishka Shah	AI-assisted listing suggestions integrated	Week 11
Nishka Shah	Listing module fully tested	Week 13
Heer Mehta	AI service scaffolding ready	Week 6
Heer Mehta	Matching engine and OCR duplicate detection functional	Week 10
Heer Mehta	AI recommendation accuracy validated	Week 13
Heer Shah	Search and filtering APIs live	Week 7
Heer Shah	In-app chat and real-time notifications live	Week 12
Heer Shah	Chat and notification services fully tested	Week 13
Charmi Bhayani	Rental APIs and QR handover module live	Week 8
Charmi Bhayani	Rental reminder scheduling live	Week 11
Charmi Bhayani	Rental module fully tested	Week 13
Kajal Varlani	Admin APIs (approval, dispute handling) live	Week 9
Kajal Varlani	Admin dashboard and analytics live	Week 11
Kajal Varlani	Admin workflows fully tested	Week 13

Team Member	Key Milestone	Target Week
Deep Boghara	Database schema and CI/version-control workflow set up	Week 2
Deep Boghara	Deployment pipeline ready	Week 12
Deep Boghara	Final deployment executed across all services	Week 14

Dependencies & Handoffs

The schedule above is shaped by a small number of key handoffs between members. These are the dates that matter most for staying on track, since a delay in any of them affects the modules that depend on it:

From	To	Handoff	Week
Deep Boghara	All members	Database schema and Git/CI workflow ready for integration	Week 4
Dhrumil Doshi	Nishka Shah, Charmi Bhayani, Kajal Varlani	Auth APIs and RBAC ready for module integration	Week 5
Nishka Shah	Heer Mehta	Listings live and ready for AI matching integration	Week 9
Heer Mehta	Heer Shah	Match-event triggers ready for notification pipeline	Week 10
Charmi Bhayani	Heer Shah	Rental due-dates ready for reminder notifications	Week 10
Kajal Varlani	Dev Barot	Analytics and moderation data ready for final report	Week 12

Timeline Chart & Task Board

The timeline chart and task board referenced throughout this document are maintained in GanttProject -style and Trello respectively. The current state of each is captured below; the live board and the exported chart are also submitted separately as described in Submission Deliverables.

Phase Overview

Phase	Milestone	Week Range	Color Key
Phase 1	Planning & Requirement Analysis	Week 1-2	Planning
Phase 2	UI/UX Design	Week 3-4	Design
Phase 3	Core Backend Development	Week 5-7	Backend
Phase 4	Core Frontend Development	Week 6-9	Frontend
Phase 5	AI Feature Integration	Week 8-10	AI
Phase 6	Real-Time Features & Notifications	Week 10-11	Real-Time
Phase 7	Testing & QA	Week 12-13	Testing
Phase 8	Deployment & Final Presentation	Week 14	Deployment

Figure 1: Phase-to-color legend used across the Gantt chart, matching the eight phases in Team Task Timeline & Milestones.

Gantt Chart — Week-wise Task View

Dev Barot (202512069) — Project Manager & System Architect

[illegible]

Dhruvil Doshi (202512088) — Authentication & User Management Lead

[illegible]

Nishka Shah (202512035) — Marketplace & Listing Management Lead

[illegible]

Heer Mehta (202512095) — AI & Intelligent Services Lead

[illegible]

Heer Shah (202512024) — Search, Chat & Notification Lead

[illegible]

Charmi Bhayani (202512028) — Rental & Transaction Management Lead

[illegible]

Kajal Varlani (202512017) — Admin Panel & Analytics Lead

[illegible]

Deep Boghara (202512076) — Database, DevOps & Quality Assurance Lead

Team Member	Task	Phase	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
Deep Boghara	DB schema & MongoDB Atlas setup	Planning														
Deep Boghara	Schema review vs UI/UX	Design														
Deep Boghara	Set up CI pipeline & automated build checks	Backend														
Deep Boghara	Support integration & version control	Backend														
Deep Boghara	Monitor performance & prep deployment pipeline	Deployment														
Deep Boghara	Testing & security checks	Testing														
Deep Boghara	Support architecture & release-readiness review	Deployment														
Deep Boghara	Manage deployment	Deployment														

Task Board — Trello

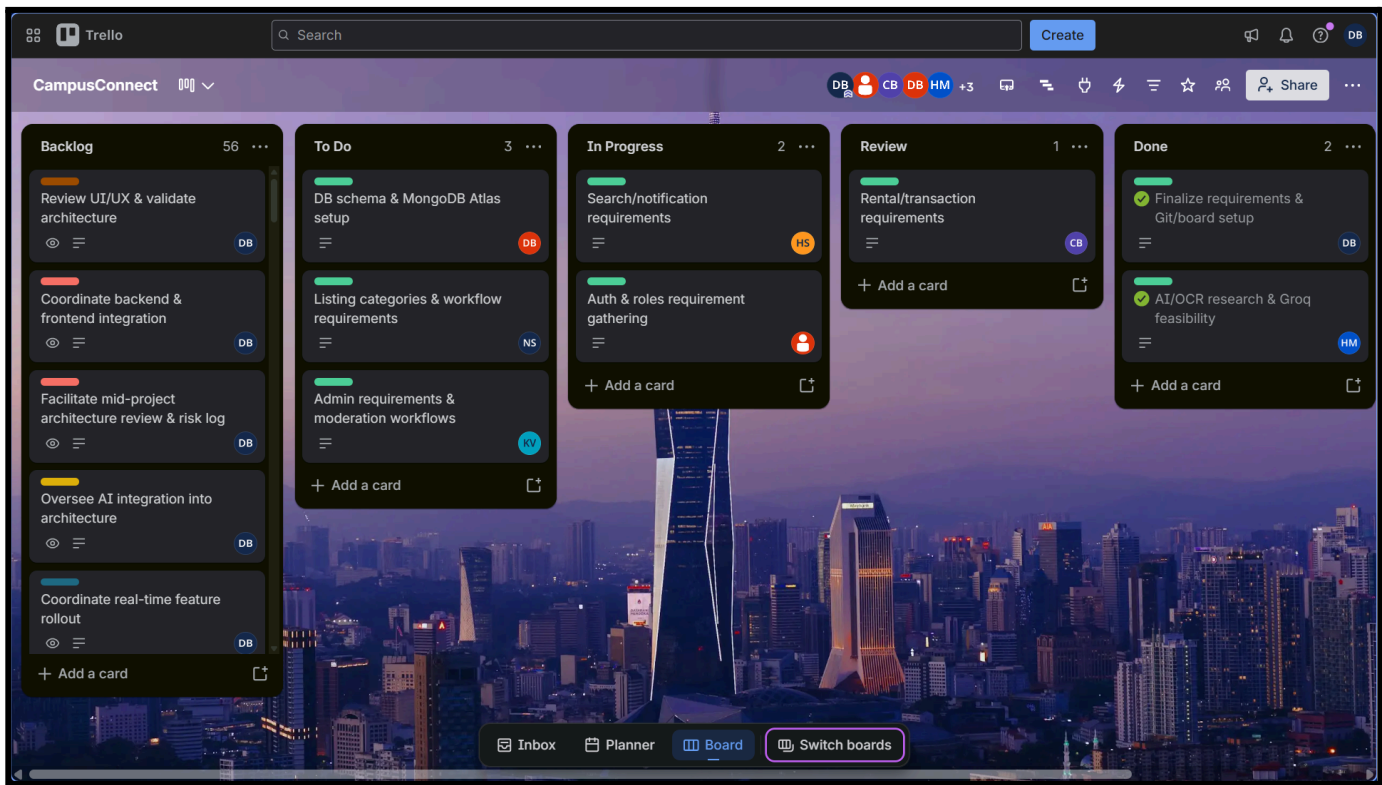


Figure 3: Current state of the CampusConnect Trello board. Backlog holds tasks not yet started; To Do, In Progress, and Review reflect work now underway following the completed planning phase; Done holds the Week 1-2 requirement-gathering tasks completed so far.